

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

20 September 2026

Abstract

Collective phase order can coordinate distributed cortex, but does not establish shared content or self-representation. We distinguish coherence, compatibility and accurate reconstruction. Shared content requires independently identified quantities on overlapping regions; compatible local descriptions determine a unique global state. A contraction fixes that state uniquely if it reconstructs the state exactly; fidelity across other states separately requires discriminating codes. Counterexamples show why a phase resultant and fixed-point uniqueness cannot replace these measurements. Lean 4 verifies the conditional implications and a model jointly satisfies their eight connecting hypotheses. This establishes mathematical consistency, not cortical realization. A declared actuator's stored mode energy bounds mean coupling and yields a necessary condition for a coherent stationary branch; cortical mode prices remain unknown. We propose a held-out comparison of field-score and structural-connectivity onset models during anaesthetic emergence. A predictive advantage would support the specified spatial model; causal field identification requires intervention. Unity and a minimal self remain interpretations without an independent measure of experience.

1 Introduction

Endogenous extracellular fields can affect spike timing and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012; Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent decoding evidence.

The familiar noisy mean-field oscillator supplies a stationary coherence threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource condition for a declared actuator, bounds that connect phase order to content only through specified encoders, and conditions under which compatible descriptions glue and an accurate reconstruction fixes the glued state. A Lean 4 development checks these implications and a joint witness satisfies their eight connecting hypotheses in a specified model. It proves neither that

23 cortex satisfies those hypotheses nor that the glued state is experienced. The
24 latter identification motivates the framework and needs its own endpoint.

25 The empirical question is which measurements could support the pro-
26 posed connections. A phase resultant can miss a locally ordered winding field;
27 stationary phase distributions can leave the coupling mechanism unidenti-
28 fied; and coherent adaptation can lose alignment with a designated content
29 template. Independent decoders must identify shared quantities before their
30 agreement can be tested. We therefore distinguish stationary branches from
31 finite observed onset, and compare prespecified field and connectivity onset
32 models on held-out emergence recordings. The comparison tests a specified
33 spatial prediction, not the presence of experience.

34 Sections 2–5 develop the three requirements; Section 6 applies their mea-
35 surement obligations to a digital candidate. Section 7 states the resource and
36 composition premises, and Sections 8–9 examine observables and the cortical
37 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
38 gives their derivations and controls.

39 **2 Limits of phase order**

40 Several limits hold without the cortical hypothesis. Equal phases can coexist
41 with incompatible local descriptions; phase order constrains shared content
42 only through encoders related on overlaps. A winding field can have zero
43 global resultant while remaining locally ordered. A constant readout can

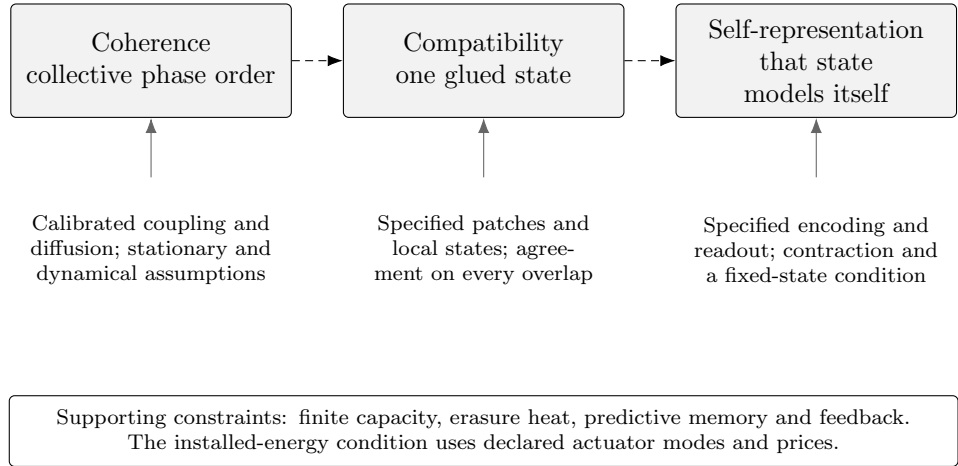


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

44 have a unique fixed point without reconstructing its input. A region that
 45 does not distinguish relevant states cannot encode them, and a site cannot
 46 report a change outside its causal past by a declared deadline. These lim-
 47 its reflect familiar phase oscillator and information constraints (Acebrón et
 48 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
 49 The sections below state each one beside the condition or measurement it
 50 motivates.

3 Coherence: physical coupling and its re- source condition

3.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency.

At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz & Mirollo, 1991). Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

3.2 Installed energy bounds mean coupling

A spatial model replaces the finite sum by integration against a kernel. One declared implementation carries response modes with continuous profiles ϕ_i , installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

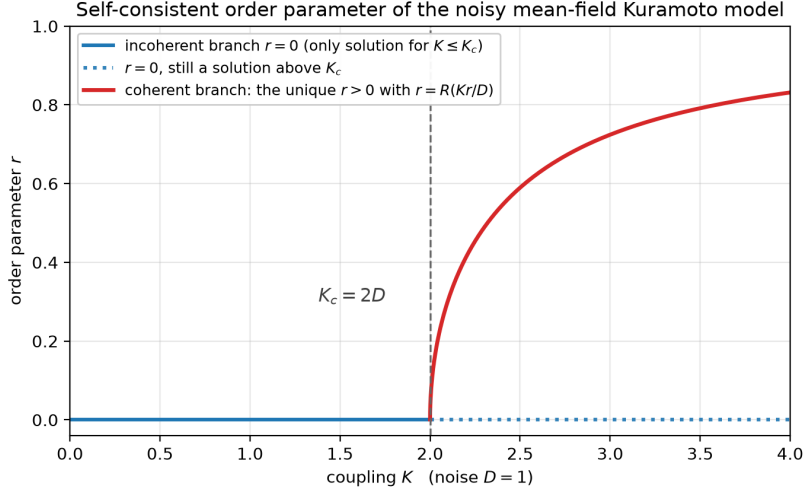


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

83 The same configuration sets the reciprocal kernel and stored energy. An
 84 executed transition changes the kernel through its changed occupancies and
 85 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 86 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 87 pair reconstruction on refining spatial partitions converges to its product-
 88 measure integral under the specified compactness and sampling assumptions;
 89 the refinement index is independent of the number of control updates.

90 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 91 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

92 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned}
 \tag{5}$$

93 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 94 coherent stationary branch and any growing Fourier mode about the uniform
 95 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 96 is necessary: energy above it need not produce sufficient coupling because
 97 the bound can be loose.

98 The theorem concerns stored energy in the declared actuator modes. The
 99 estimated cortical signalling expenditure and extracellular electrostatic en-
 100 ergy in Supplemental Material measure different quantities; neither identifies
 101 the occupancies, mode prices or their relation to metabolic supply. Conse-
 102 quently the bound supplies a necessary condition for a declared actuator,
 103 but no cortical value of κ or demonstration that cortex meets it. Applying
 104 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 105 a justified reduction.

106 4 Compatibility: from coordinated activity 107 to a global description

108 4.1 Why common timing is insufficient

109 For overlapping patches U and V , compatibility means that restricting either
110 local description to $U \cap V$ gives the same object. Descriptions can agree on
111 shared quantities while contributing different information elsewhere. Com-
112 patible sections of the specified sheaf have a unique global extension. On
113 a finite cover, local finite spatial measures agreeing as measures on every
114 overlap also have a unique finite-measure extension whose sheaf image is the
115 glued section. No mass-one normalization is imposed.

116 In the cup example, visual and reaching descriptions must express a
117 shared quantity, such as cup location, in common coordinates. Restriction
118 retains that quantity while dropping details particular to each description.
119 Agreement permits surface appearance and hand configuration to contribute
120 to one situation. The mathematical counterexample assigns the same phase
121 to every patch while two local mass profiles disagree on a shared site. Those
122 profiles have no common extension. Because they are chosen independently
123 of phase, synchrony places no constraint on their disagreement.

124 This distinction limits the biological interpretation. Receptive territories
125 and content domains must be related to spatial patches, and the descriptions
126 must support the required extension. Matching overlap marginals of de-

127 coded probability laws need not determine a unique global joint distribution
 128 (Abramsky & Brandenburger, 2011).

129 The cover is itself a choice, and the construction constrains it in one way
 130 only: synchronized patches agree where they overlap. Where they do not
 131 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 132 satisfies that obligation for *every* assignment of local descriptions and phases
 133 and the gluing theorem still returns a unique global section, which is the
 134 tuple it was handed. Membership of the class then reports nothing about the
 135 system beyond the patches covering it. What a gluing claim asks of a cover
 136 is therefore a non-degeneracy the theorem does not assume: that distinct
 137 patches meet, and that what they share is a quantity fixed independently of
 138 the agreement it is being used to establish.

139 4.2 When coherence does constrain content

140 A declared relation between phase and content gives a positive result. Let
 141 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 142 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 143 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 144 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 145 sites; their equality on overlaps is a supplied relation between descriptions of
 146 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 147 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 148 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.

149 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (6)$$

150 Here discrepancy is the norm of the difference, and r and N are read on
151 whichever population the bound is applied to. That choice is what decides
152 whether the estimate says anything. A patch carries an order parameter
153 of its own, and reading the bound there carries that patch's resultant and
154 that patch's site count, so a large population observed through small patches
155 is constrained by the small count. Read instead on the whole array, the
156 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
157 from an average over N sites. The order cannot be improved in general:
158 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
159 observations agree exactly at either scale. A shared scalar encoder is a special
160 case. Dropping the overlap relation permits disagreement even at perfect
161 locking. A common nonexpansive linear projection preserves agreement, but
162 agreement of a projection does not establish agreement of the full vector.

163 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
164 constrains anything only where the right-hand side falls below that. At
165 the sheet's own patch order, the bound is informative for at most 10 sites;
166 beyond that, the coherence-to-content link requires the bridge assumption
167 (E78) of the cortical hypothesis. Supplemental Material provides the detailed
168 derivation of this saturation, and states three conditional derivations that

169 replace the uniform factor $\sqrt{N}$ by a count of disagreeing pairs, a chain across
 170 a cover's shared sites, or a spectral gap at a frequency-locked configuration.

171 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 172 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 173 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 174 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

175 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 176 duce an initial mismatch. Approximate observations leave a nondecaying
 177 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 178 update to the phase process.

179 4.3 What approximate agreement determines

180 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 181 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 182 Weights that vanish outside their patches and sum to one at each site select
 183 a global profile by weighted averaging. The selection lies within ε of each
 184 patch there, and any two such selections differ by at most ε , independently
 185 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 186 $L\varepsilon$ across selections.

187 At zero tolerance, the selection is the unique exact extension. At every
 188 positive tolerance there are families with no exact extension; selection then

189 gives an approximation whose dependence on weights is bounded. Supple-
 190 mental Material gives a three-site witness attaining the bound. The mass-
 191 profile construction uses the scalar case of the content dynamics and addi-
 192 tionally requires nonnegativity of the initial profiles and observations. The
 193 general vector result supplies no identification with the spatial-measure sheaf.
 194 These constructions concern declared objects: a specified cover, contents
 195 valued in a normed space, and encoders related on overlaps. They select
 196 neither the neural cover nor the decoder, and identifying their contents with
 197 perceptual content, or establishing that the identification bears on experi-
 198 ence, is a further modelling task; Section 8 sets out what a measurement of
 199 it would have to establish.

200 **5 Self-representation: a state that models it-** 201 **self**

202 **5.1 The reflexive construction**

203 Compatibility gives a global state. Reflexivity relates it to an encoding within
 204 that state. Equip global sections with a complete metric. If the additional
 205 restriction-resonance condition identifies encoding with restriction to a region
 206 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

207 The region and readout are fixed across the state space. If F contracts and
 208 fixes the section s_* formed from the local descriptions, Banach's theorem
 209 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
 210 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
 211 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
 212 less than one for a relaxation time $\tau > 0$. Constructing a map with that
 213 factor remains a modelling obligation. A fixed contraction can reconstruct
 214 accurately only a restricted family of states: if its error is at most ε through-
 215 out that family, any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

216 In the cup example, the global state includes the represented scene and the
 217 perceiver's relation to it. A local encoding must retain enough information
 218 for the readout to recover that modelled state; cup position alone would be
 219 insufficient if hand configuration and viewpoint could not be reconstructed.

220 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 221 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

222 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 223 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 224 weakens.

225 5.2 Reconstruction across a family of states

226 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 227 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
 228 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

229 Thus M relevant states pairwise more than 2ε apart require at least M
 230 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 231 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 232 existing self-prediction map.

233 This is an additional accuracy criterion on a family, with no requirement
 234 for multiple exact fixed points. For the same fixed contracting map used by
 235 E89, however, the diameter bound above is an additional restriction: at zero
 236 error the family contains at most one metric-distinct state. A broad family
 237 therefore requires a different, possibly input-conditioned reconstruction map
 238 with its own accuracy and stability assumptions. A blind encoder or constant
 239 readout fails the stated criterion for separated states despite having a unique
 240 fixed point.

241 Where the encoding is read as the state's own restriction to a region, the
 242 criterion becomes a condition on the region. Restriction resonance requires
 243 the encoder and that restriction to identify the same pairs of states, so a
 244 boundary reconstructing two relevant states more than 2ε apart while its

245 region reads them identically is not resonant. Read forwards, a resonant
 246 boundary that reconstructs a pairwise separated family restricts injectively
 247 on it. Which states a region reads identically is fixed by where the region
 248 sits, before an encoding is chosen, so the resolution the family demands is
 249 the region's and no readout supplies it. The implication runs one way: a
 250 region that does resolve the family is not thereby shown to carry a resonant
 251 encoding, and E89 supplies no such resolution.

252 Resonance and resolution are jointly satisfiable without a local fold. Take
 253 the region to be the whole substrate and use the identity readout: restric-
 254 tion is the identity, so this boundary is resonant and reconstructs every de-
 255 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 256 E89's strict contraction premise on a nontrivial metric space. A properness
 257 condition on the region is therefore a separate requirement, and the fixed-
 258 contraction diameter bound remains in force when E89 is also required. With
 259 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 260 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 261 so the family is no larger than the δ -packing number of what the region car-
 262 ries. That is the shape a continuous family of states needs, and at the whole
 263 substrate it says nothing, because the region's range is everything. The gain
 264 L is declared and derived here for no physical readout, and at unbounded
 265 gain the bound says nothing.

266 **5.3 Agreement after a change takes communication**

267 Snapshot compatibility does not specify how quickly descriptions respond
268 when a shared quantity changes. In a synchronous network whose updates
269 read only the site’s own state and declared neighbours’ messages, two initial
270 configurations agreeing on a site’s causal past through T rounds produce the
271 same state there at round T . The deadline consequence is the one Section 2
272 states, and it follows from the reconstruction bound in Section 5.2: a report
273 from that site is a readout of a code that the two interventions leave identical.

274 A three-site delay line witnesses failure after one round and exact report-
275 ing after two; cutting the path prevents delivery at every round. A report
276 may still happen to be correct about one fixed world. The bound concerns
277 responsiveness across interventions.

278 Throughout this section *prediction* names a reconstruction map, and the
279 relevant family of states, the metric, the tolerance, the communication graph
280 and the deadline are all declared inputs, as are the represented variables and
281 the fidelity counted as accurate. Reading the map as temporal neural pre-
282 diction would require dynamics; reading the resulting code count as physical
283 memory capacity would require a hardware model, which a state-dependent
284 readout would have to extend.

285 6 What the conditions constrain in a digital 286 candidate

287 The results of Section 2 hold for any candidate, and for a large language
288 model (LLM) running on graphics processing units (GPUs) four of them are
289 checkable rather than merely stateable. Those four are where this framework
290 bears on claims made for such systems. It does not exclude them, and no re-
291 sult here could; what it does is fix what a claim of unity or self-representation
292 commits its maker to measuring.

293 **A quantitative bound for a linear bottleneck.** Let a finite spatial
294 coupling operator K have orthonormal eigenmodes e_i with $|\lambda_1| \geq \dots \geq |\lambda_N|$.
295 Every linear approximation A of rank at most d satisfies

$$\sum_{i=1}^N \|Ke_i - Ae_i\|^2 \geq \sum_{i=d+1}^N \lambda_i^2. \quad (11)$$

296 Retaining the first d modes attains equality. A linear encoder and readout
297 factoring through d real coordinates therefore pay this error. Applying the
298 bound requires a spectrum for the target and a rank constraint on the actual
299 approximation. Query/key dimension alone supplies no such constraint on
300 softmax attention: even a two-position rank-one logit matrix can yield rank-
301 two attention. The result prices an explicitly linear bottleneck in a finite
302 representation of a spatial kernel; it gives no universal gap between trans-

303 formers and continuous fields. Supplemental Material derives the bound and
304 checks the softmax counterexample.

305 A fixed synthetic task makes the scope visible. Two local views share
306 scene and self coordinates; the target is the global state and a self-relative
307 coordinate. On the parity-held-out family, a rank-two linear decoder’s input-
308 weighted predicted training-error floor is 2.000, and its held-out error is 2.000.
309 Equal numerical values do not make the training-law floor a held-out guaran-
310 tee. On a separate scene-family shift, the fitted full linear decoder has train-
311 ing error 0.000 and held-out error 2.667. The task’s declared target map
312 reconstructs the noiseless held-out inputs with error 0.000; the restricted
313 training family leaves one direction unidentified. A small causal attention
314 decoder and a coupled phase code also lose held-out accuracy on that shift
315 under their stated CPU fits; neither is governed by the linear rank floor. The
316 measured codes, overlap perturbations, self-state interventions and commu-
317 nication deadlines are described in Supplemental Material. These task results
318 rank no architecture family and supply no hardware-energy or experiential
319 claim.

320 **Codes, not reports.** Reconstruction across a declared family requires the
321 encoder to distinguish states more than 2ε apart when the target error is
322 ε (Eq. (10)). Verbal reports alone do not establish that condition. For
323 a proposed self-model, declare the family and fidelity, identify the internal
324 code under intervention, then test whether it separates the family. A token

sequence is one possible code; an activation region is another and requires its own measurement.

The same interface has an information ceiling. For any finite joint law of an internal state X and a length- k token sequence Y over vocabulary V , $I(X; Y) \leq H(Y) \leq k \log_2 |V|$ bits. The proof uses the full joint law and allows dependent emissions and zero-probability outcomes. A binary copy attains one bit per emitted binary symbol; an independent output conveys zero. This counts the declared output sequence, so a cache, timing signal or other accessible state must be included separately when it forms part of the channel.

The region, not the readout. If the code is a restriction of the candidate’s state to a region, that region must distinguish the relevant family before a readout is chosen. In one forward pass of a decoder-only transformer, the causal mask restricts position t to positions at or before t ; an intervention outside that causal past cannot be reconstructed from its activation alone. Across passes, prior output adds a separate channel. The bounded-fan-in communication result of Section 5 applies to a declared physical interconnect and deadline; a full-support attention graph does not satisfy its sparse-neighbourhood premise. Supplemental Material gives the graph bounds and their hardware scope.

Agreement about what. Unity requires the candidate’s local descriptions to agree on independently identified shared quantities (Section 4). Common

347 timing, a shared residual stream and fluent output supply none of that, for
348 the reason Section 2 gives: coordination bounds content disagreement only
349 through encoders already related on overlaps. Which quantities two parts
350 of a model describe in common is a decoding question, carrying the failure
351 modes of Section 8. The cover is a choice, and choosing it to secure agreement
352 empties the claim.

353 **Physical scope.** Digital representations can satisfy compatibility and re-
354 construction. The coupling-gated route additionally requires a calibrated
355 physical coupling, its noise regime and an independently identified relation
356 from phase order to shared content. An attention matrix or a shared clock
357 supplies none of these; the cortical proposal carries the same physical obliga-
358 tions. Finite-resolution capacity depends on code-space volume, resolution
359 and readout gain, so continuity alone favours no substrate. Supplemental
360 Material gives the capacity and hardware-support comparisons. Neither the
361 installed-energy condition nor those comparisons exclude a digital realization.

362 The conditions impose measurements on a digital candidate but settle
363 no question of machine experience. Supplemental Material places this com-
364 parison beside biological naturalism, substrate-dependent computation and
365 current proposals about LLM organization.

366 7 Thermodynamics and composition

367 The phase and representation results concern coupling, content disagreement
 368 and reconstruction error. A finite change in phase order has a probability-
 369 current cost bound under the conditions below. Thermodynamic heat re-
 370 quires a separate physical identification of the current and reservoir. The
 371 installed-energy condition of Section 3.2 is necessary for its declared actu-
 372 ator; whether it restricts a physical candidate depends on measured mode
 373 prices, coupling reduction and diffusion. No composition edge derives co-
 374 herence, compatibility or reconstruction from a heat bound. Finite-register
 375 capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system-
 376 bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate
 377 boundary result concerns a field reaching disconnected vacuum components
 378 (Kibble, 1976). Table 1 states the connecting assumptions.

379 7.1 Changing order carries a probability-current cost

380 For a smooth positive normalized periodic phase density ρ satisfying $\partial_t \rho =$
 381 $-\partial_\theta J$, with diffusion $D > 0$, define $\sigma_J = \int_{-\pi}^{\pi} J^2 / (D\rho) d\theta$ and $r = |\int e^{i\theta} \rho d\theta|$.
 382 Differentiation under the integral gives $\dot{r}^2 \leq D\sigma_J(1-r^2)$ wherever $0 < r < 1$.
 383 A smooth path staying in that interval therefore obeys

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau}. \quad (12)$$

Endpoint order, duration, diffusion and the probability current determine this floor. In an exploratory finite-volume onset calculation, a threshold chosen after inspecting pilot runs gives current cost 0.120 versus a floor of 0.046. These numbers describe the discretized model, not thermal or metabolic power.

The current $J = v\rho - D\partial_\theta\rho$ also satisfies

$$\left(\int_{-\pi}^{\pi} J d\theta\right)^2 \leq D\sigma_J. \quad (13)$$

A uniform density under constant drive ω attains $\sigma_J = \omega^2/D$. The identical-frequency gradient model has positive self-consistent stationary order above $K = 2D$ with $J = 0$ and $\sigma_J = 0$. Stationary order therefore has no positive current-cost floor. Nor does the bound price an externally chosen $\dot{K}$: in the identical-frequency zero-drive model, the uniform density remains an exact zero-current solution under any coupling schedule. A cortical power claim needs the physical current, reservoir and thermal conversion to be identified. This current functional is distinct from the substrate squared-drift objective used in the plasticity controls.

7.2 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (14)$$

402 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 403 et al., 2012). Zero retained information also satisfies it, so a task and a
 404 learning rule require additional specification. Relating this bound to Lan-
 405 dauer heat requires a common thermal scale and budget, and memory waste
 406 bounded by erased entropy.

407 An acting agent changes the signal it will receive. Specify policy, world,
 408 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{15}$$

409 The second conditional term vanishes in the passive model. With feedback,
 410 the signed difference can be negative as action creates correlations with the
 411 future. A causal reading requires an environmental state sufficient to describe
 412 the relevant dynamics.

413 For a controlled substep, the active thermodynamic bound instead uses
 414 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 415 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
 416 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
 417 Ito and Sagawa, 2013). Every forward path of positive probability must have
 418 a positive-probability reverse for this finite identity. A missing reverse gives
 419 infinite divergence. The supplement develops the support conditions and
 420 composes actuation with memory updates on their shared intermediate law.

421 **7.3 Learning and repeated operation need explicit re-** 422 **sources**

423 The finite models in Supplemental Material separate policy selection, learning
424 from a phase-derived sensor and repeated operation. A binary closed-loop
425 witness lets a register install the phase energy landscape, lets the phase evolve
426 under a bath-supported channel, senses the new phase and updates that same
427 register. A fixed decoder trained on one family predicts the next phase on
428 a held-out family with accuracy 0.783 under an informative sensor, versus
429 0.552 under a matched blind sensor. Their marginal phase order is the same.
430 The executed law also charges installation, sensing, reset and preparation
431 against a finite work store; phase-flip bath heat is recorded separately. The
432 symbol task and mode prices are supplied toy inputs, so this does not identify
433 a cortical mechanism or a heat cost for the full sensing and reset protocol.

434 For a complete cycle with a common energy observable $\mathcal{E}$ and thermal
435 scale $k_B T$, positive forward and reverse path support, and local detailed
436 balance at each stage, the conditional joint-law budget is

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (16)$$

437 when expected cumulative work stays within the supplied allowance B at
438 every prefix. This bounds expectations. The numerical witness specifies a
439 bath for the phase transition; its sensing, reset and preparation prices do
440 not establish reverse protocols or heat for the complete cycle. Separate con-

441 instructions track a store on each trajectory and specify when a mean allowance
 442 represents an available resource; an expected budget alone permits overdrawn
 443 paths.

444 For the declared process, a finite source with positive net cost funds
 445 finitely many cycles. A common physical ledger for the sensing and reset
 446 channels, and a cortical calibration of every mode price, remain separate
 447 obligations.

448 **7.4 What the composition establishes**

449 The passive and feedback composition theorems each accept eight connecting
 450 implications (Table 1). In the active branch, a named substep at the register’s
 451 thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (17)$$

452 with B the register’s heat. The upper allocation follows from a ledger whose
 453 operations deliver that total heat to one reservoir and whose other opera-
 454 tions do not increase joint entropy. Those ledger assumptions are physical
 455 inputs. Neither Landauer’s lower bound nor microscopic reversibility sup-
 456 plies them. Finite two-bit witnesses exhibit positive forward and reverse
 457 elementary paths, local detailed balance at one thermal scale, and interac-
 458 tion energies of order $k_B T$; these conditions alone do not quantify proximity
 459 to equilibrium. Cortex has not been shown to satisfy these premises. The

460 illustrative cortical metabolic expenditure over one region and one phase-
 461 diffusion time is 2.4×10^{14} times kT at body temperature, but identifying it
 462 with the register heat budget B requires a common physical ledger. Read-
 463 ing the register at the scale where its premises are physical, such as a single
 464 channel, requires E45Active to carry that step’s budget to the convergence
 465 of a continuum kernel. The active branch is an alternative route to the same
 466 node rather than a stronger one, and no result of Section 3.2 passes through
 467 it: the installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.

Table 1 (continued)

Connection	Status	Required assumption
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and read-out. They conclude that the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

483 in one place and machine-checked together, and not that coherence follows
484 from installed energy or self-representation from coherence.

485 Joint witnesses combine a register, boundary, refining mesh and small
486 spatial cover; active witnesses add controlled thermal updates and a priced
487 installation. They establish satisfiability under their stated identifications.
488 Mesh convergence follows from refinement data, and the witnesses establish
489 no common cortical mechanism. Supplemental Material maps the claims to
490 their Lean declarations and records the counterexamples. The development
491 declares no additional axioms beyond its libraries' foundational axioms.

492 8 What the observations establish

493 The composition holds under its premises. Whether observation can estab-
494 lish those premises is a separate question, and the three requirements it
495 connects concern different objects: phase dynamics, local descriptions and
496 internal encodings. Their observables must retain these distinctions. The
497 following controls show which mechanisms remain indistinguishable, which
498 temporal approximations fail, which states one observable cannot separate,
499 and what independent content measurements must establish. Each is a finite
500 numerical regime on a declared model, with full parameters and controls in
501 Supplemental Material, and none of them calibrates a cortical quantity.

502 8.1 Stationary distributions leave the mechanism un- 503 determined

504 Equation (2) gives a concentration–coherence relation with no fitted shape
505 parameter. Estimate r from circular resultant length and a from density
506 shape: inverting R from the measured r would make agreement automatic.
507 Even independently estimated agreement identifies a statistical family shared
508 by different mechanisms.

509 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
510 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
511 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
512 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
513 and diffusion D_0 have that same density. Their stationary distributions co-
514 incide window by window despite different causes.

515 Resolved dynamics supply additional information. In the ideal model with
516 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
517 increment means constrain drift. Separating endogenous coupling from a
518 matched common drive additionally requires a measured drive or controlled
519 perturbation; identifying electromagnetic coupling requires evidence specific
520 to that interaction. Supplemental Material gives the equivalence construction
521 and its numerical check.

522 8.2 Finite ramps change the observable onset

523 The square-root law in Eq. (3) describes stationary solutions. A changing
524 coupling requires time for the phase distribution to approach those solutions.
525 In a deterministic mean-field control that crosses the threshold symmetrically,
526 none of the scanned rates meets the declared tracking tolerance: slowing
527 the crossing does not remove the terminal residual under that initialization.
528 A slow control variable therefore supplies no general guarantee of branch
529 tracking at onset. Supplemental Material gives the residual, rate range and
530 admissible rates away from threshold.

531 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
532 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
533 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
534 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
535 6 uncensored legs supply the delay scaling below; the decade-spaced ones
536 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
537 onset exponents differ from those of the exact stationary branch evaluated
538 on the same grids. At the fastest speed the ensemble mean never reaches the
539 fit window, so its fitted slope does not estimate stationary onset.

540 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
541 makes the apparent exponent depend on ramp speed and fitting window.
542 The specified deviation-based critical ramp speed is not bracketed by this
543 sweep. With fluctuating coupling, the relevant stationary comparison also
544 depends on rate: the mean coupling approximates order in the tested fast

545 regime, while averaging stationary order over coupling works for the tested
 546 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 547 neither approximation.

548 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 549 $v^{0.444}$. The interval from fitting those means is $[0.394, 0.494]$; resampling
 550 replicas independently within each speed gives a 95% interval $[0.398, 0.496]$,
 551 with fraction 0.019 of 10000 bootstrap fits at or above $1/2$. This fraction
 552 describes the resampling distribution under the fitted protocol; it is not a
 553 test of the ideal exponent. A deterministic mean-field run seeded at the
 554 fluctuation floor gives exponent 0.425 with the same estimator and escape
 555 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls
 556 show that the criterion affects the fitted exponent, but do not prove that it
 557 accounts for the entire finite-population shortfall. The larger-population low-
 558 threshold fit has substantial log residuals; matched-speed fits do not converge
 559 toward one-half across the tested sizes and criteria. Some trajectories also
 560 cross the absolute escape level before K_c , so the first postcritical crossing can
 561 yield a zero or grid-limited delay. This finite sweep establishes no common
 562 asymptotic exponent.

563 The ideal delayed-onset calculation motivates a square-root scaling hy-
 564 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 565 tify the coupling medium. A within-subject emergence design can vary the
 566 propofol taper and test how an operational onset changes with that manipu-

567 lation, but interpreting the result as a coupling-rate law requires a measured
 568 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 569 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 570 describes how D could be estimated from resolved phase traces.

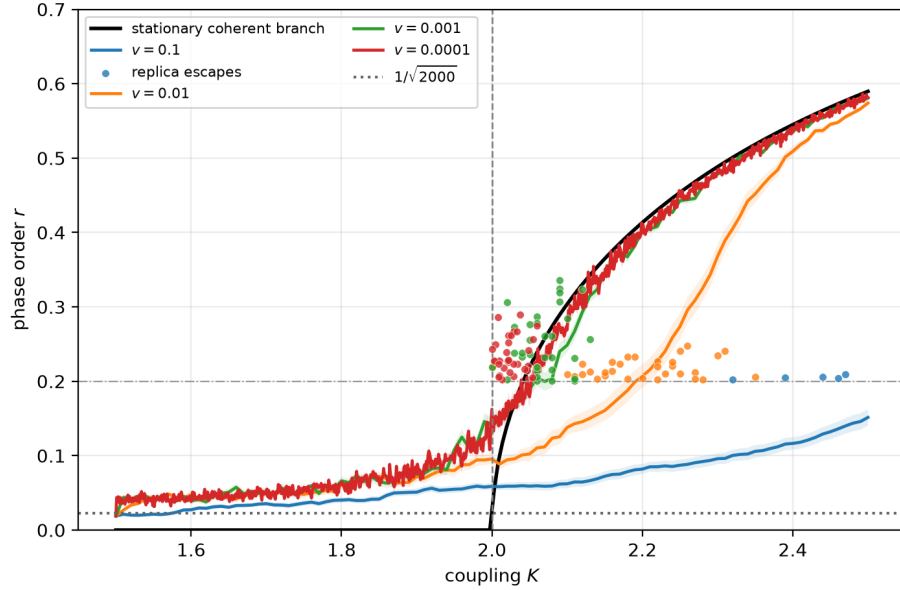


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica's first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

571 Additional controls address integration bias, population size and spatial
 572 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 573 $[1.958, 2.009]$ contains 2.

574 8.3 A winding state leaves the order parameter am- 575 biguous

576 The global order parameter r discards spatial arrangement. On a periodic
577 sheet, a phase field can wind once around the sheet and have $r = 0$ while
578 neighbouring sites remain locked. At identical frequencies, every such wind-
579 ing is stationary for an isotropic symmetric kernel: contributions at opposite
580 separations cancel. It violates the quarter-turn initial-spread premise of the
581 convergence result, which is why that result can still select the uniform state.

582 In the spatial control, the winding persists below the interpolated de-
583 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
584 and winding starts under the same coupling, diffusion and kernel end at
585 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
586 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
587 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
588 realizations. These are finite controls, not a cortical wave identification or a
589 derived transition.

590 Linearization about a winding reweights each separation's coupling by
591 the cosine of its phase advance. The resulting symmetric kernel has explicit
592 mode rates: nonnegative reweighted coupling gives no growing mode, while
593 a positive rate destabilizes that winding. The criterion concerns linear rates,
594 not asymptotic stability, and the measured retention boundary need not
595 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$

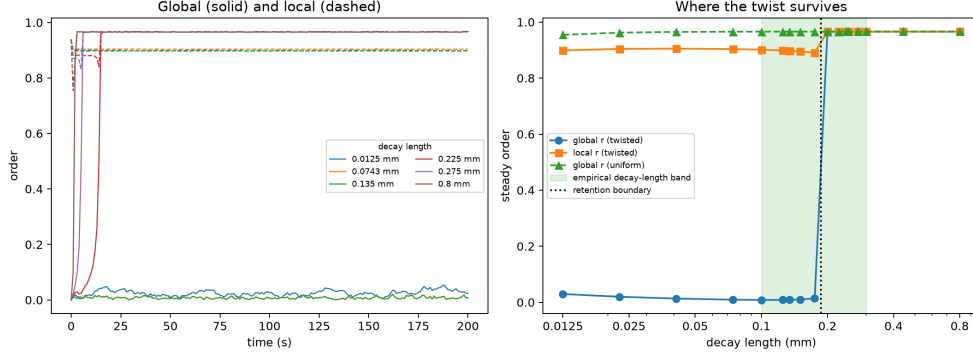


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

596 0, where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 597 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 598 winding from that model even though the phase-only equations retain it as
 599 a stationary state.

600 Neither model turns these states into measured cortical spirals. The pe-
 601 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 602 no propagation speed, and the amplitude variable has no calibrated relation
 603 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 604 and defect results. Locally, Eq. (6) can still bound content disagreement on a
 605 winding patch when shared encoders are supplied, although its global $r = 0$

form is uninformative. A nonzero ring winding number q also obstructs a global real-valued phase lift; it has no identified content meaning here. Thus an onset criterion based only on r can register a change of basin as well as a change of coupling. The protocol of Section 9.3 needs a spatial phase observable.

8.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters.

The formalization proves descent of the squared-drift objective for a specified gradient flow at fixed phases. Its functional differs from thermodynamic entropy production: at a coherent stationary density the probability current vanishes while phase-averaged squared drift is positive. Supplemental Material gives the operator, its gradient and the conditions of the descent result.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. Its template is a diagnostic covariance

of latent signals, never a training input. Across three illustrative seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these

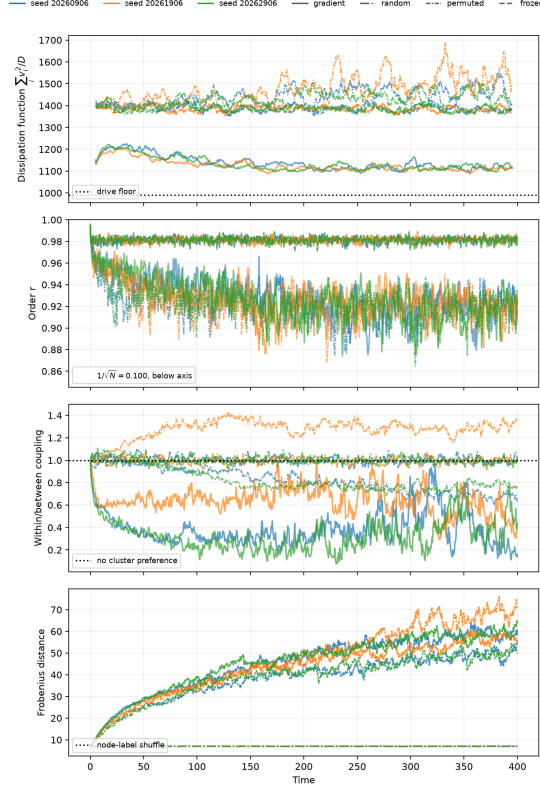


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed content, phase carries a share 1.000 of the content entropy when content is built from it and 0.000 when it is not. A phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. The implemented content statistic detects their incompatibility. These controls validate a measurement on known constructions; a neural content model and decoder remain to be established.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. The useful choice must therefore be constrained by what neural activity encodes. The exact composition assumes compatibility of its supplied cover. The content mechanisms above identify additional hypotheses under which agreement can arise; they leave the empirical choice of content and the connection to that cover explicit.

8.6 Estimator calibration on exploratory EEG

The concentration range and the dependence structure decide whether the concentration-coherence relation can be tested at all. Under the specified

671 log-density estimator and bias-floor criterion, discrimination at the observed
672 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
673 designs, rising to 31000 when sites share a mean in groups of 10. These
674 counts describe that estimator and dependence model; the intended recording
675 requires its own calibration. Supplemental Material reports the bin-count
676 controls and the full dependence-aware design.

677 That ceiling comes from exercising the estimators on public propofol-
678 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms win-
679 dows. Bipolar and common-reference montages produce different residuals.
680 Bipolar-montage scalp EEG sits in the linear Bessel regime ($a_{\max} = 0.542$,
681 deviation from $a/2 < 1\%$). The exercise confirms the estimator is well-
682 behaved in this regime but does not test the nonlinear prediction. The
683 bipolar results are compatible with the Bessel relation and with monotone
684 alternatives that agree to this precision over the sampled range. Temporal
685 pooling changes the estimand, and sedation recordings leave the awakening
686 hypothesis untested.

687 An informative dataset therefore needs sufficient concentration range or
688 precision, calibrated estimators and an observation model for dependence,
689 source mixing and reference choice. Intracranial recordings (ECoG, sEEG or
690 LFP) or narrowband burst analysis may reach a more informative concen-
691 tration range, but that range must be measured. At the observed ceiling,
692 the specified design requires 3000 independent sites or 31000 sites under its
693 clustered dependence model; these are design calculations, not sample re-

quirements transferable to another recording. Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used here. Site count, state coverage and bandwidth are separate requirements.

9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what observations must distinguish before a phase trace can support this identification.

9.1 Physical identification and calibration

The cortical hypothesis identifies part of effective coupling with endogenous extracellular electromagnetic interactions. Fields can modulate spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models propose coordination of distributed neural information (Pinotsis & Miller, 2022; McFadden, 2020). Phase organization under propofol motivates studying this regime (Bastos et al., 2021), but does not measure coupling.

The formal field predicate requires a jointly continuous kernel, a normal-

713 ized substrate, coupling equal to the kernel average, and no single site pro-
714 viding more than half that strength. Toy constructions satisfy it. A cortical
715 realization must specify the scale at which the kernel and its reduction ap-
716 ply: calcium imaging reports different state-dependent modularity patterns
717 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

718 Cortical columns are the obvious candidate unit, and they answer one
719 of the two demands the formalism makes. As the oscillator index they are
720 a serious proposal: they are delineated by response properties and anatomy
721 rather than by the agreement they would be used to establish (Mountcastle,
722 1997), and a phase can be estimated from a column-sized territory at the
723 electrode spacings used for cortical wave measurements (Townsend et al.,
724 2015). As the content cover they are a poor one, for three separate reasons.
725 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
726 the compatibility obligation without constraining anything (Section 4). The
727 cover condition additionally requires the patches to exhaust the substrate,
728 while much of association cortex has no accepted columnar parcellation. And
729 the column’s standing as a canonical cortical unit is contested (Horton &
730 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
731 supplies the oscillator index; the content cover remains an open empirical
732 question, which the bridge assumption E78 records.

733 The spatial reduction is also testable. A row-sum aggregation at zero
734 diagonal tracks the uniform-coupling control over the tested spatial decay
735 range, while frequency heterogeneity displaces the operational threshold. The

736 fit window biases both estimates relative to $2D$; agreement with the control
 737 is therefore the supported conclusion. Supplemental Material reports the
 738 reduction, its numerical error and the distinct assumptions needed for the
 739 kernel-average identification in Eq. (5).

740 The field commitment requires physical evidence beyond a choice of math-
 741 ematical representation, and the available support comparisons do not supply
 742 it. A finite matrix spread over positive-mass cells preserves its total cou-
 743 pling and phase correlation exactly, while point-supported vanishing under
 744 an atomless measure and a disadvantage from excluding a better-correlated
 745 pair concern specified support constraints; Section 6 reads the same compar-
 746 isons against a digital candidate. Neither supplies energy prices.

747 A spike-time shift alone supplies no coupling rate. The dimensionless
 748 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 749 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (18)$$

750 This rate describes how induced timing shifts accumulate into drift. Nei-
 751 ther it nor the installation factor κ follows from a heat bound. Testing the
 752 physical threshold requires independent estimates of the interaction kernel,
 753 its aggregation and diffusion in the same cortical regime.

754 9.2 Extracellular geometry and recovery

755 The proposed control variable is extracellular geometry. Extracellular vol-
756 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
757 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
758 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
759 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
760 an effective-medium description predicts that a larger extracellular fraction
761 raises conductivity and reduces field amplitude.

762 The coupling-gated hypothesis consequently predicts that extracellular
763 contraction on awakening raises E and, holding the other factors in Eq. (18)
764 fixed, raises effective coupling. This argument supplies a direction; the
765 geometry–coupling relation, its magnitude and its role in human recovery
766 require measurement.

767 For a smooth crossing at nonzero speed, evaluating the stationary branch
768 along the changing control gives a deterministic tracking approximation that
769 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (19)$$

770 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
771 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
772 proximation under the tested initialization. A separation of geometry and
773 phase time scales can support tracking away from threshold. The awakening

774 hypothesis needs a calibrated dynamical prediction for an operational onset
775 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
776 recovery latency.

777 **9.3 A protocol for the awakening experiment**

778 Neural inertia describes a behavioural barrier and hysteresis between anaes-
779 thetic induction and emergence (Friedman et al., 2010); recovery can pass
780 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
781 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
782 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
783 mechanism from a synaptic one. A possible discriminator is spatial onset
784 relative to independently estimated local field coupling and network connec-
785 tivity. The present mean-field model has no spatially varying κ and derives
786 no nucleation order, so a field-based onset map needs a separate model spec-
787 ified before the observations are inspected.

788 For that comparison, partition the recording into prespecified regions i
789 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
790 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
791 and a connectivity score c_i from a separately estimated structural network.
792 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
793 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

794 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (20)$$

795 Predicted onset is the first crossing of a region-specific threshold fixed from
796 pre-onset fluctuations. Fit the two scales on training sessions, then com-
797 pare held-out onset times and site order under the same observation model.
798 Larger field scores predict earlier onset only when baseline order and diffu-
799 sion do not reverse the ranking. This declared approximation omits interre-
800 gional coupling and saturation; a successful ranking would support its field
801 covariate, not prove ephaptic causation. Supplemental Material specifies the
802 controls and failure conditions.

803 A within-subject protocol can manipulate emergence rate, with spatially
804 resolved recordings to compare the proposed field and connectivity models:

- 805 1. **Manipulate emergence rate.** Vary the propofol infusion taper
806 across sessions within the same subject, recording drug concentrations
807 or a validated pharmacokinetic proxy to quantify the manipulation.
- 808 2. **Measure behavioural recovery.** Record time-to-response to a stan-
809 dard stimulus as the independent behavioural endpoint, rather than
810 estimating ΔK directly.
- 811 3. **Resolve spatial onset.** Identify anatomical locations where coher-
812 ence first crosses the baseline-scaled threshold and score the two maps

813 of Eq. (20) on held-out sessions. The two covariate maps are fixed
814 before the outcome sessions are inspected.

815 **4. Match physical controls.** Confirm that the observed coherence-
816 to-amplitude relation holds under the specific local diffusion D and
817 observation noise, estimable from the quadratic variation of resolved
818 phase traces.

819 Predeclare the recording window, regional fluctuation thresholds, and a
820 shared timing-error model. Fit the two scales and that observation model
821 on training sessions only; on held-out sessions, score the observed site-onset
822 times and sites that never cross by the same regional predictive rule. Report
823 site order as a secondary outcome and behavioural time-to-response sepa-
824 rately. Sessions without a crossing are censored observations, not zero delays
825 or exclusions. Compare the two scores only where independently estimated
826 field and connectivity maps predict distinguishable onset patterns at the
827 recording resolution. A held-out field-model advantage would support that
828 specified spatial predictor; it would not identify an ephaptic cause, shared
829 content or experience. Supplemental Material gives the controls and pilot
830 quantities needed to judge whether the comparison is informative.

831 **10 Discussion**

832 **10.1 Unity and a minimal self as interpretive motiva-** 833 **tion**

834 We propose the state glued from compatible local descriptions as a formal
835 correlate of conscious unity, and its accurate internal reconstruction as a cor-
836 relate of a minimal self. In the cup example, this would mean one description
837 that preserves the cup’s relation to the perceiver as well as the scene. Com-
838 patibility permits distinct contents and uncertainty; the shared quantities
839 must be chosen from independent evidence about what each region encodes.
840 The reconstruction need not be a verbal or autobiographical account. Its
841 proposed relation to experienced perspective remains an interpretation. Nei-
842 ther identification is among the eight hypotheses in Table 1, and satisfying
843 them would not measure experience.

844 **10.2 What a test of the full proposal must establish**

845 A test needs separate estimates of effective coupling and diffusion, shared
846 content on overlapping regions, and reconstruction from a specified encoding
847 at a declared fidelity. The cover, relevant state family and decoder must
848 be fixed without selecting them for agreement. An independent experiential
849 endpoint is needed to evaluate the interpretation; absence of a report alone
850 does not define absence of experience. A unified episode with incompati-

851 ble decoded descriptions would challenge the unity identification. A unified
852 episode lacking the proposed self-encoding would challenge the minimal-self
853 claim. Conversely, satisfying the mathematical conditions without such an
854 episode would challenge their sufficiency as correlates.

855 For the physical route, cortical actuator modes and prices, the geometry–
856 coupling relation and a dynamical onset model remain to be measured. The
857 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
858 diction. Controlled changes in coordination and independently decoded con-
859 tent agreement would address the representational connection. The synthetic
860 decoder and EEG exercises calibrate parts of these measurements; they do
861 not identify a neural content code or test experience.

862 **10.3 Explanatory scope**

863 Integrated-information, global-workspace and predictive-processing accounts
864 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
865 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
866 relations among coupling, local content and reconstruction. Its formal field
867 condition requires a continuous kernel, a normalized substrate, a specified
868 average coupling and noise, and a site-contribution bound. Those conditions
869 do not identify an electromagnetic substrate; the extracellular interpretation
870 needs the separate calibration of Section 9. The joint witness shows that
871 the formal hypotheses can hold together in a declared model. A cortical
872 realization and an experiential interpretation require additional evidence.

873 The remaining explanatory question is whether these relations occur in
874 neural activity and matter for experience. The framework identifies candidate
875 correlates and tests of their physical and representational premises. It does
876 not explain why satisfying them would be accompanied by experience.

877 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances

Table 2 (continued)

Symbol	Defined	Meaning
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (17)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (18)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s \mathcal{A})$
$\mathcal{E}$	Eq. (16)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (18)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

878 **Data and code availability**

879 The Lean development, simulation scripts, tests and saved numerical
880 summaries are maintained in the GitHub repository `bobaseb/coupling_
881 gated_cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and
882 Mathlib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publi-
883 cation simulation values are generated from saved summaries without re-
884 running production sweeps. The exploratory EEG analysis uses Open-
885 Neuro dataset `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022,
886 1033, 1045, 1056, resting awake and sedation runs 1–2); preprocessing
887 and selection are described in Supplemental Material and implemented in
888 `simulations/empirical_collapse.py`. The repository contains the repro-
889 ducible analysis and does not redistribute the raw recordings.

890 **Funding and competing interests**

891 This research received no external funding. The author declares no compet-
892 ing interests.

893 **Use of AI tools**

894 AI assistance to this work was substantial and is not separable by task. Sev-
895 eral coding assistants and chat interfaces were used in an interleaved way
896 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,

897 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
 898 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
 899 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
 900 accessed with the Hermes CLI. Between them they contributed to the Lean
 901 development, the simulation and analysis code, and the drafting and revision
 902 of both documents. Because the tools were used in combination and in alter-
 903 nation, no section, proof, figure or result is attributable to any one of them,
 904 and none is claimed to be. The author is responsible for all content, including
 905 everything a tool produced. Machine checking establishes the stated Lean
 906 results under their hypotheses; it does not validate biological interpretations
 907 or replace scientific review of generated code and prose.

908 **References**

- 909 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
 910 electric fields emerge from and allow representational drift. *NeuroImage*,
 911 253, 119058.
- 912 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
 913 tion theory: from consciousness to its physical substrate. *Nature Reviews*
 914 *Neuroscience*, 17(7), 450-461.
- 915 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
 916 Mind. *Oxford University Press*.

- 917 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
 918 tational theory of conscious processing. *Current Opinion in Neurobiology*,
 919 25, 76-84.
- 920 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
 921 *Reviews Neuroscience*, 11(2), 127-138.
- 922 Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness:
 923 A case for biological computationalism. *Neuroscience & Biobehavioral Re-*
 924 *views*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.
- 925 Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial
 926 consciousness through the lens of neuroscience. *Trends in Neurosciences*,
 927 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.
- 928 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
 929 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- 930 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
 931 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
 932 doi:10.1017/S0140525X26106906.
- 933 Gurnee, W., et al. (2026). Verbalizable Representations Form a
 934 Global Workspace in Language Models. *Transformer Circuits Thread*.
 935 arXiv:2607.15495.
- 936 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-

937 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
938 40, 24-50.

939 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
940 parts of the free energy principle. *Entropy*, 23(3), 293.

941 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
942 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

943 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
944 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

945 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
946 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
947 *Progress in Neurobiology*, 226, 102465.

948 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
949 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
950 9(8), 45-60.

951 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
952 testable hypothesis about the characteristics of conscious as opposed to
953 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

954 McFadden, J. (2020). Integrating information in the brain’s EM field: the
955 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
956 niaa016.

- 957 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
958 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 959 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
960 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
961 95-103.
- 962 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
963 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 964 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
965 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 966 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-
967 logical Reviews*, 98(1), 239-389.
- 968 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
969 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.
- 970 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
971 without a function. *Philosophical Transactions of the Royal Society B*,
972 360(1456), 837-862. doi:10.1098/rstb.2005.1623.
- 973 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National
974 Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.
- 975 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-

976 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
 977 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

978 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
 979 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
 980 ity of complexity measures to capture dreaming during propofol sedation.
 981 *Scientific Reports*, 15, 32746.

982 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
 983 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
 984 brain. *Science*, 342(6156), 373-377.

985 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
 986 *Physiological Reviews*, 88(4), 1277-1340.

987 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
 988 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

989 Logothetis, N. K., Kayser, C., & Oeltermann, A. (2007). In vivo measure-
 990 ment of cortical impedance spectrum in monkeys: implications for signal
 991 propagation. *Neuron*, 55(5), 809–823. doi:10.1016/j.neuron.2007.07.027.

992 Gabriel, S., Lau, R. W., & Gabriel, C. (1996). The dielectric properties of bio-
 993 logical tissues: II. Measurements in the frequency range 10 Hz to 20 GHz.
 994 *Physics in Medicine and Biology*, 41(11), 2251–2269. doi:10.1088/0031-
 995 9155/41/11/002.

- 996 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
997 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
998 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- 999 Landauer, R. (1961). Irreversibility and heat generation in the computing
1000 process. *IBM Journal of Research and Development*, 5(3), 183-191.
- 1001 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1002 *of Physics A: Mathematical and General*, 9(8), 1387-1398.
- 1003 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1004 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- 1005 Strogatz, S. H., & Mirollo, R. E. (1991). Stability of incoherence in a popu-
1006 lation of coupled oscillators. *Journal of Statistical Physics*, 63, 613-635.
- 1007 Acebrón, J. A., Bonilla, L. L., Pérez Vicente, C. J., Ritort, F., &
1008 Spigler, R. (2005). The Kuramoto model: A simple paradigm for
1009 synchronization phenomena. *Reviews of Modern Physics*, 77, 137-185.
1010 doi:10.1103/RevModPhys.77.137.
- 1011 Ott, E., & Antonsen, T. M. (2008). Low dimensional behavior of
1012 large systems of globally coupled oscillators. *Chaos*, 18, 037113.
1013 doi:10.1063/1.2930766.
- 1014 Dörfler, F., & Bullo, F. (2014). Synchronization in complex net-
1015 works of phase oscillators: A survey. *Automatica*, 50, 1539-1564.
1016 doi:10.1016/j.automatica.2014.04.012.

1017 Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature*
1018 *Neuroscience*, 20, 340–352. doi:10.1038/nn.4497.

1019 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1020 of prediction. *Physical Review Letters*, 109(12), 120604.

1021 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1022 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

1023 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
1024 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
1025 031015.

1026 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
1027 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
1028 PhysRevLett.111.180603.

1029 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
1030 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

1031 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1032 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1033 181.

1034 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1035 archive.pitt.edu/115.

- 1036 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1037 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1038 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1039 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1040 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1041 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1042 Dai Pra, P., & den Hollander, F. (1996). McKean–Vlasov limit for interacting
1043 random processes in random media. *Journal of Statistical Physics*, 84, 735–
1044 772. doi:10.1007/BF02179656.
- 1045 Bertini, L., Giacomin, G., & Pakdaman, K. (2010). Dynamical aspects of
1046 mean field plane rotators and the Kuramoto model. *Journal of Statistical*
1047 *Physics*, 138, 270–290. doi:10.1007/s10955-009-9908-9.
- 1048 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
1049 bifurcations with additive noise. *Probability Theory and Related Fields*,
1050 122, 341–388. doi:10.1007/s004400100174.
- 1051 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
1052 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
1053 doi:10.1088/1367-2630/13/11/113036.
- 1054 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
1055 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

1056 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
 1057 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

1058 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
 1059 terns underlie complex brain dynamics and are related to cognitive pro-
 1060 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
 1061 023-01626-5.

1062 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
 1063 and dissipation with partially observed fluctuating signals. *Physical Review*
 1064 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

1065 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
 1066 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
 1067 cell resolution functional networks during unconsciousness are segre-
 1068 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
 1069 doi:10.1016/j.celrep.2025.116902.

1070 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
 1071 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
 1072 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.

1073 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the
 1074 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
 1075 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.

- 1076 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
1077 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.
- 1078 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.
- 1079 Baer, S. M., Erneux, T., & Rinzel, J. (1989). The slow passage through a
1080 Hopf bifurcation: delay, memory effects, and resonance. *SIAM Journal on*
1081 *Applied Mathematics*, 49(1), 55–71. doi:10.1137/0149003.
- 1082 Friedman, E. B., Sun, Y., Moore, J. T., Hung, H.-T., Meng, Q. C., Perera, P.,
1083 Joiner, W. J., Thomas, S. A., Eckenhoff, R. G., Sehgal, A., & Kelz, M. B.
1084 (2010). A conserved behavioral state barrier impedes transitions between
1085 anesthetic-induced unconsciousness and wakefulness: evidence for neural
1086 inertia. *PLoS ONE*, 5(7), e11903. doi:10.1371/journal.pone.0011903.
- 1087 Hudson, A. E., Calderon, D. P., Pfaff, D. W., & Proekt, A. (2014). Recovery
1088 of consciousness is mediated by a network of discrete metastable activity
1089 states. *Proceedings of the National Academy of Sciences*, 111(25), 9283–
1090 9288. doi:10.1073/pnas.1408296111.
- 1091 Proekt, A., & Hudson, A. E. (2018). A stochastic basis for neural inertia
1092 in emergence from general anaesthesia. *British Journal of Anaesthesia*,
1093 121(1), 86–94. doi:10.1016/j.bja.2018.02.035.